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DEVICE FOR PROVIDING BRAIDS, SYSTEM COMPRISING THE DEVICE AND METHOD FOR PROVIDING BRAIDS

Abstract

A device for providing nettings by means of a construction machine, in particular in mining and/or tunnel construction, has at least one netting receiving apparatus which is configured to receive and at least partially enclose a netting unit and which has a netting receiving opening and a netting output opening, wherein at least the netting output opening is equipped with at least one inhibiting element which is configured to prevent uncontrolled netting output.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a U.S. national stage application of international patent application PCT/EP2023/060513, filed on Apr. 21, 2023, which is based on and claims priority to German patent application DE 10 2022 110 418.4, filed on Apr. 28, 2022, the contents of which are incorporated herein by reference.

PRIOR ART

[0002] The invention concerns a device for providing nettings, a system and a method for providing nettings.

[0003] A device for providing nettings by means of a construction machine in mining and/or tunnel construction is already known from WO 2010/015339 A1.

[0004] The objective of the invention is in particular to provide a generic device having improved properties with regard to providing nettings. The objective is achieved according to the invention.

ADVANTAGES OF THE INVENTION

[0005] The invention is based on a device for providing nettings by means of a construction machine, in particular in mining and/or tunnel construction, having at least one netting receiving apparatus which is configured to receive and at least partially enclose a netting unit and which has a netting receiving opening and a netting output opening.

[0006] It is proposed that at least the netting output opening is equipped with at least one inhibiting element which is configured to prevent uncontrolled netting output. As a result, advantageous netting laying, in particular along tunnel or mining walls, can be made possible, which in particular manages without hydraulics, pneumatics and/or electronics. Controlled netting output, in particular using particularly simple means, can advantageously be achieved. Purely mechanical netting handling can advantageously be made possible. In particular, the netting receiving apparatus is configured to receive a netting unit and to provide it for processing, in particular for fastening the netting unit to a wall, for example a tunnel wall and/or a mine wall and/or an adit wall and/or a mountain wall or a comparable structure. In particular, the netting unit comprises the netting to be provided. In particular, the netting unit is configured as a netting roll. Alternatively, however, the netting unit could also be of a shape which deviates from a roll, for example as a folded netting stack or the like. In particular, the netting unit comprises at least one, preferably precisely one, netting web. In this case, the netting web is preferably substantially longer than wide. Typical lengths of netting webs are, for example, 15 m, 20 m, 25 m or 30 m. Typical widths of netting webs are, for example, 2 m, 2.5 m, 2.7 m or 3 m. In particular, the netting is designed as a steel wire netting, preferably as a steel mesh wire netting. In particular, the netting is designed from at least one high-tensile steel wire. In this case, a “high-tensile steel wire” should be understood to mean, in particular, a steel wire having a tensile strength of at least 800 N/mm², preferably of at least 1200 N/mm², advantageously of at least 1770 N/mm², preferably of at least 2200 N/mm² and particularly preferably of at most 3000 N/mm². In particular, the netting forms a safety net which, in the installed state, is configured to prevent a rockfall or the like and/or to reduce the effects thereof. In particular, the netting is formed from a plurality of flat helices which are braided into one another and of which at least one helix is produced from at least one individual wire, a wire bundle, a wire strand, a wire rope and/or another longitudinal element having at least one wire. However, alternative types of netting, such as hexagonal nettings or ring nettings, etc., are also conceivable. Preferably, the device is configured for providing what are known as TECCO™ nettings from Geobrugg AG (Romanshorn, Switzerland), in particular MINAX™ nettings (e.g., MINAX 80/4 or MINAX 80/3) from Geobrugg AG. “Configured” should be understood to mean, in particular, specially programmed, designed and/or equipped. The fact that an object is configured for a specific function should be understood to mean, in particular, that

the object performs and/or carries out this specific function in at least one application state and/or operating state.

[0007] The construction machine can be designed as a tunnel construction machine, preferably as a bolter or as a jumbo, or else as a separate machine which is specifically configured for holding the netting receiving apparatus. In particular, the construction machine has at least one boom. In particular, the netting receiving apparatus is configured to be mounted on the boom, in particular on one end of the boom. In particular, the netting receiving apparatus is configured to be moved along a tunnel wall or the like by means of the boom of the construction machine. In particular, the netting receiving apparatus can be aligned in any desired position and/or location with respect to the wall in order to provide the nettings. In particular, the netting receiving apparatus provides the netting for extraction. In this case, the netting can be extractable manually or with aids, for example a second boom of the construction machine which is designed as a drilling arm. In particular, the netting receiving apparatus has a shape which allows at least one netting unit to be received, in particular inserted and/or suspended. In particular, the netting receiving apparatus encloses the at least one netting unit, in particular a netting unit which is configured as a netting roll, at least partially, preferably at least to a large extent, at least in the circumferential direction. In this context, a large part should be understood to mean, in particular, 60%, preferably 75%, preferably 90% and particularly preferably 95%. For this purpose, the netting receiving apparatus can have, for example, an at least substantially cylindrical shape. In this case, it is conceivable for cover surfaces of the cylindrical shape to be of closed or open or at least openable design. If the cover surfaces are of openable design, a lateral introduction of netting units would be conceivable. However, an arrangement of the netting receiving opening on the circumference of the cylindrical shape is preferred.

[0008] In particular, the netting receiving apparatus is formed from a metallic material, for example from a steel. In particular, the netting receiving apparatus is formed from a structurally rigid, in particular flexurally rigid, material. In particular, the material of the netting receiving apparatus is selected in such a way that even a falling back-and-forth of the netting unit in the netting receiving apparatus during an installation movement of the netting receiving apparatus cannot cause any damage, such as bendings, etc. In particular, the netting receiving apparatus has a weight of less than 200 kg, preferably of less than 150 kg and preferably of less than 100 kg. In particular, the netting receiving apparatus has a load-bearing capacity for netting units of more than 500 kg, preferably of more than 750 kg and preferably of more than 1000 kg. In particular, the netting receiving apparatus is configured at least for receiving netting rolls having a length of approximately 2.7 m, a diameter of approximately 0.7 m and/or a weight of approximately 210 kg. In particular, the netting receiving apparatus forms an unrolling unit for steel wire nettings provided in a rolled-up state. In particular, the netting receiving apparatus has a center of gravity which lies in a vicinity of the boom of the construction machine and/or of a mounting point of the netting receiving apparatus for mounting on the construction machine. In this case, the term “vicinity” should be understood to mean, in particular, a region formed from points, wherein each point lies at a distance of at most 0.5 m, preferably at most 0.3 m, from the boom of the construction machine and/or from the mounting point of the netting receiving apparatus. In particular, the netting receiving apparatus can be configured as a type of cage. Alternative implementations, for example having at least partially closed walls, are conceivable.

[0009] In particular, the netting receiving opening is arranged on a radial circumference of the netting receiving apparatus, in particular on a cylinder sheat of the cylindrical shape. In particular, the netting receiving opening has an elongate shape. In particular, an extension of the netting receiving opening in the axial direction of the netting receiving apparatus is at least twice as large, preferably at least three times as large and particularly preferably at least four times as large as an extension in a direction perpendicular to the axial direction of the netting receiving apparatus (circumferential direction). In particular, the netting receiving opening has dimensions which allow

the netting unit to be inserted easily into a netting receiving space of the netting receiving apparatus, for example at least $2.5\text{ m} \times 1\text{ m}$. In particular, the netting receiving opening is designed as a recess and/or opening in the netting receiving apparatus, wherein the opening and/or recess can be formed as the non-enclosing part of the netting receiving apparatus. In particular, the netting receiving opening could be at least partially closed. In particular, the open state could be configured to allow the netting unit to be inserted and/or removed. In particular, the closed state could be configured to hold the netting unit in the netting receiving apparatus, in particular to prevent it from falling out. In particular, the netting dispensing opening is arranged on a radial circumference of the netting receiving apparatus, in particular on a cylindrical sheath of the cylindrical shape. In particular, the netting dispensing opening has an elongate shape. In particular, the netting dispensing opening has a slot-like design. In particular, an extension of the netting dispensing opening in the axial direction of the netting receiving apparatus is at least four times as large, preferably at least seven times as large and particularly preferably at least ten times as large as a further extension in a direction perpendicular to the axial direction of the netting receiving apparatus (circumferential direction). In particular, the netting receiving opening has dimensions which allow the netting of the netting unit to be dispensed areally from the netting receiving space of the netting receiving apparatus, for example at least $2.5\text{ m} \times 0.2\text{ m}$. In particular, the netting dispensing opening is formed by the netting receiving opening. In particular, the netting dispensing opening is designed to be smaller than the netting receiving opening. In particular, the netting dispensing opening is designed as a part of the netting receiving opening.

[0010] In particular, the netting of the netting unit is provided by the netting dispensing opening. In particular, the netting unit which is inserted into the netting receiving apparatus is partially unrolled for the purpose of being provided and is guided through the netting dispensing opening. In particular, a netting unit which is inserted into the netting receiving apparatus can be extracted from the netting dispensing opening. In particular, the inhibiting element is configured to prevent automatic unrolling/automatic dispensing of the netting, for example by gravity. In particular, the inhibiting element is configured to hold the netting in a substantially fixed unrolled state/extracted state. In particular, the inhibiting element is designed differently from the netting receiving apparatus. In particular, the inhibiting element is formed from a different material than the netting receiving apparatus. In particular, the inhibiting element is configured to provide the netting in a controlled and/or controllable manner. It is conceivable for the netting receiving apparatus to have a plurality of inhibiting elements, in particular two, three or more than three inhibiting elements. In particular, the number and/or implementation of the inhibiting elements can depend on the size and/or the weight of the netting unit, and also further properties of the netting unit and/or of the netting receiving apparatus. In particular, the inhibiting element forms a type of lip of the netting dispensing opening. In particular, the inhibiting element sweeps over the netting from at least one side during the dispensing of the netting. In particular, the inhibiting element produces fixing of the netting in a current dispensing position, for example by means of a frictional force. In particular, the inhibiting element clamps the netting to the netting receiving apparatus or to a further inhibiting element.

[0011] Furthermore, it is proposed that the inhibiting element is of elastic design. As a result, simple, preferably hydraulic-, pneumatic- and/or electronic-free, inhibition of the netting can advantageously be achieved. Different types of netting can advantageously be inhibited. Damaging of the netting unit during netting output can advantageously be at least substantially avoided and/or prevented. A strength of the inhibition by the inhibiting element can advantageously be set precisely, for example by a choice of elasticity. In particular, the inhibiting element is formed from an elastic material, in particular a plastic, a rubber, a metallic material, such as, for example, a (high-tensile) steel sheet or plates/sheets made of other ferrous or nonferrous metals (e.g. an aluminum block), and/or a wood material, such as, for example, an elastic chipboard and/or a particularly elastic type of wood, such as ash, hickory or elm. It is also conceivable for the

inhibiting element to be formed from a composite material, for example a type of brake pad, as is known in particular from vehicle construction, and/or a (composite) nonwoven. It is also conceivable for the inhibiting element to be formed as a (metallic) spring which is configured to generate a pressing, in particular on account of the spring-elastic properties of the spring element, on the netting unit. The inhibiting element is preferably formed as a plate and/or slotted plate. It is additionally conceivable for the inhibiting element to be formed in the shape of a brush and/or a block and/or a ring and/or a disk. Alternatively, the inhibiting element can also be formed from a elasticoelastic material (cf. joint sealing material). In particular, the inhibiting element can be deflected flexibly out of a rest position. In particular, the inhibiting element is deflected by the netting during netting output. In particular, the inhibiting element is in the rest position when the netting dispensing opening is free of a netting. In particular, the inhibiting element is in an elastically deflected position/state when a netting is guided out of the netting dispensing opening. [0012] Furthermore, it is proposed that the inhibiting element is configured to apply a pressing to at least part of the netting. As a result of the pressing, netting output can advantageously be controlled and/or suppressed in a particularly simple, preferably hydraulic-, pneumatic- and/or electronic-free manner. A particularly high ease of use of the device for providing nettings can advantageously be achieved. Simple handling of netting units can advantageously be made possible. In particular, the pressing generated by the inhibiting element presses the netting against an edge of the netting dispensing opening. Alternatively, the pressing can also press the netting against an opposite inhibiting element. In particular, the inhibiting element applies the pressing to one of the two flat sides of the netting. In particular, the inhibiting element applies the pressing to an inner flat side of the netting or to an outer flat side of the netting. In particular, the inner flat side of the netting is designed as the side which, in the rolled-up state in a netting roll, faces a roll-up axis of the netting roll. In particular, the outer flat side of the netting is designed as the side which, in the rolled-up state in a netting roll, faces away from the roll-up axis of the netting roll. In particular, a plurality of inhibiting elements can be arranged for instance along a common line. Alternatively, an offset between individual, for example between adjacent, inhibiting elements can also be provided. In particular, the inhibiting elements are arranged without overlap with respect to one another. Alternatively, however, an at least partial overlap of adjacent inhibiting elements is also conceivable.

[0013] Furthermore, it is proposed that the inhibiting element is configured to apply the pressing to at least that part of the netting which is currently emerging from the netting output opening. As a result, effective inhibition for preventing self-rolling can advantageously be achieved. Fixing of the netting close to an emergence point from the netting receiving apparatus can advantageously be achieved, as a result of which, in particular, handling of the netting during application to a surface can be simplified. In particular, the inhibiting element is arranged in a vicinity of the netting dispensing opening, in particular of an edge of the netting dispensing opening. In this case, a “vicinity” should be understood to mean, in particular, a region which is formed by points which are at a distance of at most 25 cm, preferably at most 15 cm and preferably at most 7 cm, from an edge of the netting dispensing opening. Alternatively or additionally, however, it is also conceivable that an inhibiting element is arranged in an interior of the netting receiving apparatus and exerts a pressing on a still rolled-up part of a netting roll.

[0014] In addition, it is proposed that the inhibiting element is arranged on the netting receiving apparatus in such a way that, at least in an unloaded state, the inhibiting element covers the netting dispensing opening at least in sections at least for the most part. As a result, effective inhibition for preventing self-rolling can advantageously be achieved, in particular in a technically particularly simple and/or cost-effective manner. In particular, the inhibiting element is arranged on the netting receiving apparatus in such a way that, when the netting emerges from the netting dispensing opening, the netting must elastically deform the inhibiting element, for example bend away from the netting dispensing opening. In the unloaded state, the inhibiting element preferably completely

overlaps the netting dispensing opening at least in a spatial direction, in particular in a circumferential direction around the netting receiving apparatus. In particular, in the unloaded state, the netting receiving apparatus is empty or is equipped with a netting roll which has not been pulled out. In particular, the inhibiting element still partially overlaps/covers the netting dispensing opening even when the netting emerges from the netting dispensing opening. In particular, the inhibiting element only in sections overlaps/covers the netting dispensing opening in the axial direction of the netting receiving apparatus, in particular in the unloaded state and in a loaded state of the inhibiting element. In particular, the inhibiting element can be arranged on an upper side of the netting receiving opening, on a lower side of the netting receiving opening or on both opposite sides of the netting receiving opening. In principle, as an alternative or in addition, inhibiting elements which press laterally against the netting could also be provided.

[0015] If the inhibiting element is replaceably attached to the netting receiving apparatus, repair and/or maintenance can advantageously take place in a very simple manner. A high ease of use can advantageously be achieved. The inhibiting element can, for example, be screwed to the netting receiving apparatus or clamped to the netting receiving apparatus or plugged onto the netting receiving apparatus, for example by means of a snap closure and/or a quick-release fastener. In particular, the inhibiting element is non-destructively replaceable. In particular, the inhibiting element forms a wear part.

[0016] Furthermore, it is proposed that the netting receiving apparatus has at least one, in particular movably supported, closure element for at least partially closing the netting receiving opening. As a result, secure reception of the netting unit can advantageously be achieved. In addition, simple filling of the netting receiving apparatus can advantageously be made possible. In particular, the closure element closes the netting receiving opening at least to such an extent that the netting unit cannot fall out of the netting receiving apparatus. In particular, the closure element only partially closes the netting receiving opening, in particular in the axial direction of the netting receiving apparatus and/or in the circumferential direction of the netting receiving apparatus. The closure element preferably closes the netting receiving opening in the circumferential direction by at least 80%, preferably by at least 90%. The closure element preferably closes the netting receiving opening in the axial direction by at most 25%, preferably by at most 15%. In particular, the closure element forms a closure slide, a cover and/or a flap. In particular, the closure element is formed from an at least substantially identical material to a remaining part of the netting receiving apparatus forming the netting receiving space.

[0017] If, in this case, the netting receiving apparatus has at least one guide rail which is configured to guide the closure element on a circumference of the netting receiving apparatus, a compact design and/or comfortable handling can advantageously be made possible. In addition, incorrect operation can advantageously be prevented.

[0018] If, in addition, the closure element in a closed state delimits the netting dispensing opening in sections, an overall weight of the netting receiving apparatus can advantageously be kept low. In addition, in particular manual operation of the closure element can advantageously remain easy to handle. In particular, the closure element has a grip element for manually adjusting the position of the closure element relative to the netting receiving apparatus.

[0019] Furthermore, it is proposed that the closure element, in particular a positioning of the closure element along the guide rail, is configured to adjust the pressing applied by the inhibiting element. As a result, a high degree of flexibility can advantageously be achieved. In addition, a high degree of readiness for use can advantageously be achieved, in particular by simple and/or rapid readjustment being made possible. In addition, a simple design can advantageously be achieved. In particular, a size of the netting dispensing opening in the circumferential direction can be achieved by adjusting a setting position of the closure element, as a result of which, in particular, a degree of a necessary bending of the inhibiting element for the netting dispensing opening to pass through a netting can be adjusted. In particular, the greater the elastic

bending/deflection of the inhibiting element by the netting, the greater the pressing generated. In particular, a partially worn inhibiting element can be readjusted, in particular retensioned, by the positioning of the closure element.

[0020] In addition, it is proposed that the closure element is designed to be substantially narrower than the netting receiving opening and/or the netting dispensing opening in a main extension direction of the netting receiving opening and/or the netting dispensing opening. Material costs can advantageously be reduced. Particularly advantageously a weight of the netting receiving apparatus can be reduced. In this context, “substantially narrower” should be understood to mean, in particular, at least 50% narrower, preferably at least 70% narrower and preferably at least 85% narrower. A width of the closure element is preferably approximately one ninth of a width of the netting receiving opening. In particular, the netting receiving opening extends completely between two opposite axial end regions of the netting receiving apparatus.

[0021] Furthermore, it is proposed that the inhibiting element is arranged at a netting output opening-side end of the closure element. As a result, a simple design can advantageously be achieved. A particularly simple adjustment of the pressing force generated by the inhibiting element can advantageously be achieved. In particular, the inhibiting element extends beyond the netting output opening-side end of the closure element. In particular, the inhibiting element forms a netting output opening-side extension of the closure element.

[0022] Furthermore, it is proposed that the inhibiting element and/or the closure element has a width which corresponds to at least one mesh extent, preferably at least two mesh extents, of the netting configured for provision. As a result, the device can advantageously be operated in a trouble-free manner, in particular since the inhibiting element and the netting unit cannot become entangled. In addition, a particularly uniform inhibiting effect can advantageously be achieved over an entire web length of the netting unit. In particular, the width is an extension of the inhibiting element and/or of the closure element parallel to the axial direction of the netting receiving apparatus. In particular in the case of a mesh wire netting, such as, for example, the TECCO™ netting, the mesh extent is defined by a distance between two adjacent bending regions of a helix of the mesh wire netting and/or between two contact points of adjacent helices of the mesh wire netting. Customary distances of this type (i.e. customary minimum widths of inhibiting elements/closure elements) are 45 mm, 65 mm or 80 mm. The widths of the closure elements and of the inhibiting elements fastened thereto are preferably approximately the same size. Alternatively, however, a closure element can also carry a plurality of inhibiting elements arranged next to one another.

[0023] Furthermore, it is proposed that the netting receiving apparatus has at least one further closure element which is in particular at least substantially identical to the closure element. Secure reception of the netting unit can advantageously be achieved while maintaining as low a weight as possible and/or while maintaining as simple a handling as possible. In particular, three or more than three closure elements are also conceivable. In particular, the closure elements are arranged spaced apart from one another along the axial direction. It is conceivable for two or more closure elements to be connected to one another by means of a connecting element in order to advantageously allow the closure elements to be closed and opened easily and/or uniformly. Alternatively, however, each closure element can also be designed such that it can be operated individually. In particular, the closure elements are designed and/or dimensioned in such a way that each individual closure element or at least the combination of all closure elements is strong enough to carry a netting unit. In particular, during the movement of the device through the construction machine, the netting unit can also be displaced in such a way that its entire weight lies exclusively on the closure elements. In this case, too, secure holding of the netting unit in the netting receiving apparatus must be ensured.

[0024] If the netting receiving apparatus forms an at least substantially cylindrical netting receiving space which is configured to receive at least the netting unit, proper unrolling of netting units

which are designed as netting rolls can advantageously be achieved. In addition, a high degree of stability and/or a high degree of compactness of the netting receiving apparatus can advantageously be achieved. In particular, orientation of the netting roll within the netting receiving space does not matter with regard to operability of the device.

[0025] Furthermore, if the netting receiving apparatus has a netting receiving space which is overwidth, in particular in relation to web widths of netting units which can be provided by the netting receiving apparatus, and which is configured to provide a tolerance range for varying packing widths of, in particular otherwise identically designed, netting units, a high ease of use can advantageously be achieved. In particular, reception of netting units can advantageously be simplified, in particular even if said netting units are not rolled up and/or stacked in a perfectly straight manner. In particular, the overwidth is at least 10%, preferably at least 15% and preferably at least 20% of a web width of a largest netting unit which can be provided by the netting receiving apparatus. If the netting receiving apparatus is configured, for example, for providing approximately 2.3 m wide (MINAXTMM 80/4) netting rolls, the width of the netting receiving space is preferably approximately 2.7 m.

[0026] In addition, it is conceivable for the netting receiving apparatus to be of width-adjustable design. As a result, the dimensions of the netting receiving apparatus could advantageously be adapted to different construction machines and/or to different netting units.

[0027] If, in addition, at least one wall of the netting receiving apparatus has lightweight construction recesses, an overall weight can advantageously be kept low. In this case, the lightweight construction recesses can have different geometries, e.g. slots, holes, meshes, etc. In particular, the lightweight construction recesses are arranged distributed over at least a large part of an overall surface area of the netting receiving apparatus. In this context, a “large part” should be understood to mean, in particular, 60%, preferably 70%, preferably 80% and particularly preferably 90%.

[0028] Furthermore, it is proposed that the device has a protective unit which covers at least part of an outer side of the netting receiving apparatus in order to avoid injuries and/or soiling. A maintenance interval can advantageously be extended. A service life can advantageously be increased. Safety can advantageously be increased. In particular, the protective unit is configured as a tarpaulin, as a net, as a textile or as a comparable planar element. In particular, the protective unit is fastened to the outer side of the netting receiving apparatus. In particular, the protective unit has—if at all—substantially smaller openings than the netting receiving apparatus, with the result that dirt, for example (sprayed) mortar from tunnel construction fastening operations (“concrete rain”) cannot penetrate into the netting receiving space and/or that no fingers or other body parts can penetrate into the netting receiving space.

[0029] In addition, it is proposed that the netting receiving apparatus has a support element for positioning the netting receiving apparatus on a base. The netting receiving apparatus can advantageously be set down safely, in particular in a stable manner, as a result. The netting receiving apparatus can advantageously be positioned with respect to a base by the support element in such a way that insertion into the netting receiving space and closure of the netting receiving space can be carried out as simply as possible. In particular, the support element is formed in one piece with the netting receiving apparatus. “In one piece” should be understood to mean, in particular, cohesively connected, for example by a welding process and/or adhesive bonding process etc., and particularly advantageously integrally formed, such as by production from a casting and/or by production in a single-component or multi-component injection molding process. “In one piece” should advantageously also be understood to mean in one piece/monolithically. “In one piece” should be understood to mean, in particular, formed in one piece.

[0030] Furthermore, it is proposed that at least one region of the netting receiving apparatus which delimits the netting receiving opening has a netting insertion aid element which is in particular configured to support an insertion of the netting unit into a netting receiving space of the netting

receiving apparatus. The netting unit can advantageously be inserted and/or received into the netting receiving apparatus in a particularly simple manner, in particular even at least in a partially automated manner. In particular, the netting insertion aid element is designed as a bevel which is preferably arranged on a lower edge of the netting receiving opening in the installation direction, in particular on an edge facing the installation element. In particular, the netting insertion aid element is configured to provide the netting receiving apparatus with an excavator blade function when the netting receiving opening is open.

[0031] Furthermore, it is proposed that the netting receiving apparatus has a bearing unit for rotatably bearing netting roll central rods. A bearing of netting rolls in the netting receiving space and/or an unrolling of the netting rolls can advantageously be improved as a result. In particular, the bearing unit comprises at least two (round) receiving openings for netting roll central rods arranged opposite one another in the axial direction of the netting receiving apparatus at lateral ends of the netting receiving apparatus.

[0032] In addition, it is proposed that the netting receiving apparatus has a fastening unit for fastening to the construction machine. Simple mounting can advantageously be made possible as a result. In particular, the fastening unit is located in the vicinity of the center of gravity of the netting receiving apparatus. In particular, the fastening unit forms the mounting point of the netting receiving apparatus. In particular, the netting receiving apparatus is reinforced in the region of the fastening unit.

[0033] In addition, it is proposed that the netting receiving apparatus has stiffening elements which extend parallel to a main extension direction of the netting receiving apparatus. A design can advantageously be improved. A stiffness can advantageously be increased. A weight can advantageously be reduced.

[0034] As an alternative to a plate-like design of the inhibiting element, it is proposed that the inhibiting element is designed as a roll. Guiding of the netting output can advantageously be achieved as a result. It is additionally conceivable for the above-described elastic, e.g. plate-like, inhibiting elements to be additionally equipped, in end regions, with rolls for guiding the nettings. Protection against copying can advantageously also be achieved by the rolls, in particular in that they are designed and/or arranged in such a way that only nettings having specific mesh shapes and/or having specific mesh sizes can be reliably output from the respective netting receiving apparatus by the rolls. The rolls could advantageously not be designed such that they can be removed non-destructively from the closure elements.

[0035] Furthermore, it is proposed that the device has a plurality of inhibiting elements which are designed as rolls and are arranged in accordance with a mesh extent (mesh width) or a half mesh extent (mesh width) of the netting unit configured for provision. A type of protection against copying can advantageously be achieved as a result.

[0036] Furthermore, a system having the device and having at least one netting unit, in particular a netting roll, and having a construction machine configured for laying nettings is proposed, wherein the device is fixedly connected to the construction machine. As a result, advantageous netting laying, in particular along tunnel or mining walls, can be made possible, which in particular manages without hydraulics, pneumatics and/or electronics. In particular, the device can be rotated as desired on the boom of the construction machine, in particular through 360° and more, without the operability being impaired or the netting unit threatening to fall out of the netting receiving apparatus.

[0037] In addition, a method for providing nettings, in particular in mining and/or tunnel construction, by means of the device or by means of the system is proposed, wherein an uncontrolled netting output is prevented in at least one method step. As a result, advantageous netting laying, in particular along tunnel or mining walls, can be made possible, which in particular manages without hydraulics, pneumatics and/or electronics.

Description

DRAWINGS

[0038] Further advantages result from the following description of the drawings. Exemplary embodiments of the invention are illustrated in the drawings. The drawings, the description and the claims contain numerous features in combination. The person skilled in the art will expediently also consider the features individually and combine them to form meaningful further combinations.

[0039] In the drawings:

[0040] FIG. 1 shows a schematic illustration of a system for laying nettings in mining and/or in tunnel construction having a device, a netting unit and a construction machine,

[0041] FIG. 2 shows a schematic view of a detail of the netting of the netting unit,

[0042] FIG. 3 shows a schematic perspective view of the netting unit and the device with a netting receiving apparatus having open closure elements,

[0043] FIG. 4 shows a further schematic perspective view of the device with the netting receiving apparatus, wherein the netting unit is arranged in a netting receiving space of the netting receiving apparatus and wherein the closure elements are closed,

[0044] FIG. 5 shows a schematic vertical sectional illustration of the device with the netting receiving apparatus and with a protective unit,

[0045] FIG. 6 shows a schematic perspective rear view of the netting receiving apparatus,

[0046] FIG. 7 shows a part of a schematic vertical sectional illustration of the device with the netting receiving apparatus, the closure element of which is equipped with an alternative inhibiting element, and

[0047] FIG. 8 shows a schematic flow diagram of a method for providing nettings in mining and/or tunnel construction.

DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0048] FIG. 1 shows a schematic illustration of a system 56 for laying nettings in mining and/or in tunnel construction. The system 56 has a construction machine 10. The construction machine 10 is designed as a bolter/jumbo. The construction machine 10 is configured for drilling holes in rock walls 68. The construction machine 10 is modified in such a way that it is additionally configured for laying the nettings. The system 56 has a device 58. The device 58 is fixedly connected to the construction machine 10. The construction machine 10 has a boom 62. The device 58 is fastened to the boom 62. The device 58 is fastened to the boom 62 instead of a drill. The device 58 is fastened to the outermost end of the boom 62. The construction machine 10 has a further boom 64. The further boom 64 has a drill 66. The further boom 64 is configured for fastening the nettings to the rock wall 68. The device 58 is configured for providing the nettings for mounting on the rock wall 68. The system 56 has a netting unit 14. The netting unit 14 is configured as a netting roll. The construction machine 10 is configured to move the device 58 along the rock wall 68 and in the process to unroll the netting roll for dispensing the netting. Meanwhile, the drill 66 fastens the netting, which has already been unrolled, to the rock wall 68.

[0049] FIG. 2 shows a schematic view of a detail of the netting of the netting unit 14. The netting is designed as a wire net. The wire of the wire net is formed by a high-tensile steel. The netting is designed as a safety net for securing against small rockfall/rockfall. The netting has a plurality of helices 70. The helices 70 engage in one another in each case. The helices 70 are braided into one another. The helices 70 are aligned parallel to one another. The helices 70 each have the shape of a flattened helix. The helices 70 each have a sequence of alternating legs 72, 74. The helices 70 each have bending regions 76. A bending region 76 connects two legs 72, 74 of a helix 70 in each case. In the tensioned state of the netting, mutually engaging helices 70 touch one another in a vicinity of the bending regions 76, preferably at the bending regions 76. The legs 72, 74 each span a bending angle 78. The bending angle 78 is less than 90°. The netting forms a diamond-shaped mesh shape

80. The netting has a mesh extent **30**. The mesh extent **30** corresponds to a width of the diamond-shaped mesh shape **80** parallel to a longitudinal direction **82** of the helices **70**. The helices **70** are each formed from an individual wire. Alternatively, helices **70** could also be formed from a wire bundle, a wire strand, a wire rope and/or another longitudinal element.

[0050] FIG. **3** shows a schematic perspective view of the device **58** for providing nettings by means of the construction machine **10**. In addition, FIG. **3** schematically shows a netting unit **14** which is configured as a netting roll. The device **58** has a netting receiving apparatus **12**. The netting receiving apparatus **12** is configured to receive the netting unit **14**. The netting receiving apparatus **12** is configured to partially enclose the received netting unit **14** (see also FIG. **4**). The netting receiving apparatus **12** forms a netting receiving space **34**. The netting receiving space **34** is of approximately cylindrical shape. The netting receiving space **34** is configured to receive the netting unit **14**. The netting receiving space **34** of the netting receiving apparatus **12** is designed to be overwidth in relation to a web width **84** of a netting unit **14** which can be provided by the netting receiving apparatus **12**. As a result of the overwidth, the netting receiving space **34** provides a tolerance range for varying packing widths of otherwise identically designed netting units **14**.

[0051] The netting receiving apparatus **12** delimits the netting receiving space **34** by walls **36**, **90**. The walls **36**, **90** of the netting receiving apparatus **12**, in particular the walls **36** which delimit the receiving space in the circumferential direction **92**, have lightweight construction recesses **38**. The lightweight construction recesses **38** are configured as round holes/bores/punchings. However, other shapes are also conceivable. The netting receiving apparatus **12** has a support element **44**. The support element **44** is configured for positioning the netting receiving apparatus **12** on a base. The support element **44** forms a planar support surface and/or positioning feet on the otherwise cylindrical circumference of the netting receiving apparatus **12**. The netting receiving apparatus **12** has a bearing unit **48** for optionally rotatably bearing netting roll central rods (not illustrated)—if these are present. The bearing unit **48** is formed by holes arranged opposite one another in the axial direction **86** of the netting receiving apparatus **12** in the walls **90** which delimit the netting receiving apparatus **12** in the axial direction **86**. The netting receiving apparatus **12** has stiffening elements **54**, **88**. A part of the stiffening elements **54** extend parallel to a main extension direction **52** of the netting receiving apparatus **12**. The stiffening elements **54** which extend parallel to the main extension direction **52** of the netting receiving apparatus **12** are arranged on a wall **36** which delimits the netting receiving space **34** in the circumferential direction **92**. A further part of the stiffening elements **88** extend perpendicularly to the main extension direction **52** of the netting receiving apparatus **12**. The stiffening elements **88** which extend perpendicularly to the main extension direction **52** of the netting receiving apparatus **12** are arranged on a wall **90** which delimits the netting receiving space **34** in the axial direction **86**.

[0052] The netting receiving apparatus **12** forms a netting receiving opening **16**. The netting receiving opening **16** forms an access for introducing the netting unit **14** into the netting receiving space **34**. The netting receiving opening **16** can be partially closed. The netting receiving apparatus **12** has a closure element **22**. The closure element **22** is configured to partially close the netting receiving opening **16**. The closure element **22** is movably supported. The closure element **22** is displaceably supported in the circumferential direction **92**. The closure element **22** is manually displaceable. To simplify the manual displaceability of the closure element **22**, grip elements (not illustrated) can be provided. The netting receiving apparatus **12** has a guide rail **24**. The guide rail **24** is configured to guide the closure element **22** on the circumference of the netting receiving apparatus **12**. The guide rail **24** is covered by transverse struts **94**. The transverse struts **94** serve for captive holding of the closure element **22** in the guide rail **24**. The transverse struts **94** serve for reinforcement/stiffening of the netting receiving apparatus **12**. The closure element **22** is substantially narrower than the netting receiving opening **16** in a main extension direction **26** of the netting receiving opening **16**. The closure element **22** has a width **100** which corresponds to at least one mesh extent **30** of the netting configured for provision.

[0053] The netting receiving apparatus **12** has a further closure element **32**. The further closure element **32** is at least substantially identical to the closure element **22**. In the exemplary embodiment illustrated in the figures, the netting receiving apparatus **12** has a total of three closure elements **22, 32**, wherein more or less are also conceivable. The closure elements **22, 32** are arranged and/or designed in such a way that a netting unit **14** (see also FIG. **4**) which is arranged in the netting receiving space **34** cannot fall out of the netting receiving space **34** irrespective of a spatial position of the netting receiving apparatus **12**. In FIG. **3**, the closure elements **22, 32** are illustrated in a completely open position. The netting receiving apparatus **12** has a netting insertion aid element **46**. The netting insertion aid element **46** is arranged in a region of the netting receiving apparatus **12** which delimits the netting receiving opening **16**. The netting insertion aid element **46** is arranged in an edge region of the netting receiving opening **16** which faces the support element **44**. The netting insertion aid element **46** is configured to support an insertion of the netting unit **14** into the netting receiving space **34**. The netting insertion aid element **46** is designed as a bevel **96**. The bevel **96** tapers the netting receiving apparatus **12** towards the netting receiving opening **16**. In the position of the netting receiving apparatus **12** in which it is set up on the support element **44**, the bevel **96** forms a type of ramp for the insertion of the netting unit **14**. The netting receiving apparatus **12** can be used by the netting insertion aid element **46** in a manner of an excavator blade when the closure elements **22, 32** are open. The netting insertion aid element **46**, in particular the bevel **96**, extends over the entire axial length of the netting receiving opening **16**, in particular of the netting receiving apparatus **12**.

[0054] FIG. **4** shows a further schematic perspective view of the device **58** for providing nettings by means of the construction machine **10**, wherein the closure elements **22, 32** are positioned in a position in which they close the netting receiving opening **16**. In this state, the netting receiving apparatus **12** forms a netting dispensing opening **18** which, in particular, is of different size than the netting receiving opening **16** described above. The netting dispensing opening **18** is of slot-like design. The netting dispensing opening **18** overlaps with the netting receiving opening **16**. A lower edge (located in the direction of the support element **44**) of the netting receiving opening **16** likewise forms a lower edge of the netting dispensing opening **18**. However, an upper edge (as seen in the installation direction of the netting receiving apparatus **12**) of the netting receiving opening **16** is different from an upper edge of the netting dispensing opening **18**. The upper edge of the netting dispensing opening **18**, as seen in the installation direction of the netting receiving apparatus **12**, is formed/delimited substantially by netting output opening-side ends **28** of the closure elements **22, 32**. In a closed state, the closure elements **22, 32** directly partially delimit the netting dispensing opening **18**.

[0055] The netting receiving apparatus **12** has an inhibiting element **20**. The netting dispensing opening **18** is equipped with the inhibiting element **20**. The inhibiting element **20** is configured to prevent uncontrolled netting output of the netting unit **14** introduced into the netting receiving space **34**. The inhibiting element **20** is configured to prevent automatic unrolling of the netting roll. The inhibiting element **20** is configured to allow tensioning of the netting by a movement of the netting receiving apparatus **12** along the rock wall **68**. The inhibiting element **20** is arranged at the netting output opening-side end **28** of the closure element **22**. Here, the inhibiting element **20** is replaceably attached to the netting receiving apparatus **12**, in particular to the closure element **22**. The inhibiting element **20** has a width **98** which corresponds to at least the mesh extent **30** of the netting configured for provision. In the case illustrated, the width **98** of the inhibiting element **20** corresponds approximately to the width **100** of the closure element **22**. However, deviations in the widths **98, 100** and/or arrangements with a plurality of inhibiting elements **20** on a closure element **22** are likewise conceivable. Each of the closure elements **22, 32** is equipped with one or more inhibiting elements **20** on the netting output opening side.

[0056] The inhibiting element **20** is arranged on the netting receiving apparatus **12**, in particular the closure element **22**, in such a way that, in an unloaded state (i.e. in a state in which the netting does

not emerge through the netting dispensing opening **18**), the inhibiting element **20** completely covers the netting dispensing opening **18** in a portion which corresponds to the width **98** of the inhibiting element **20**. The unloaded closed state is not illustrated in the figures. In FIG. **4**, the netting is partially pulled out of the netting receiving space **34**, that is to say passes through the netting dispensing opening **18**. The inhibiting element **20** is of elastic design. The inhibiting element **20** is formed from a rubber material. The inhibiting element **20** is designed as a rubber plate. The elastic inhibiting element **20** is elastically bent/elastically deflected by the netting emerging from the netting dispensing opening **18** (see also FIG. **5**). The inhibiting element **20** is configured to apply a pressing to part of the netting. By virtue of the fact that the inhibiting element **20** wishes to return to its rest position/its initial shape by virtue of its elasticity, the inhibiting element **20** produces the pressing on that part of the netting which is in touch contact with the inhibiting element **20**. Consequently, the inhibiting element **20** is configured to always apply the pressing to that part of the netting which is currently emerging from the netting dispensing opening **18**. By virtue of the arrangement of the inhibiting element **20** at the netting output opening-side end **28** of the closure element **22**, the closure element **22** is configured to adjust the pressing applied by the inhibiting element **20**. The smaller the netting dispensing opening **18** is selected by the adjustment of the closure element **22**, the stronger/narrower the inhibiting element **20** has to be bent by the netting emerging from the netting dispensing opening **18**.

[0057] FIG. **5** shows a schematic vertical sectional illustration of the device **58** with the netting receiving apparatus **12**. The closure element **22** is in a closed state. The netting roll which is arranged in the netting receiving space **34** is illustrated as partially pulled out of the netting dispensing opening **18**. The inhibiting element **20** is strongly deformed/bent outward by the pulling out of the netting. The inhibiting element **20** consequently presses the netting against an opposite edge of the netting dispensing opening **18**, in particular against the netting insertion aid element **46**/the bevel **96**. The closure element **22** is fixed at an end **102** facing away from the netting output opening-side end **28** on a main body of the netting receiving apparatus **12**/on the guide rail **24**. By way of example, a bolt fixing of the closure element **22** is shown in FIG. **5**. However, alternative fixing methods are of course conceivable. The closure element **22** can be fixed in different closed positions which each bring about slightly different (vertical) sizes of the netting dispensing opening **18**. As a result, a strength of the pressing/inhibition generated by the inhibiting element **20** can be set and/or a partially worn inhibiting element **20** can be readjusted, depending on the choice of the closed position of the closure element **22**.

[0058] The device **58** has a protective unit **40**. The protective unit **40** is illustrated schematically only in FIG. **5**. In the further figures, the protective unit **40** is not shown for reasons of clarity. The protective unit **40** covers an outer side **42** of the netting receiving apparatus **12**. The protective unit **40** is provided in order to avoid injuries, such as, for example, clamping of fingers when reaching through the lightweight construction recess **38**. The protective unit **40** is provided in order to avoid penetration of soiling, such as, for example, concrete, mud or dirt, into the netting receiving space **34**. The protective unit **40** is formed by a fine-meshed net (wire net). Alternatively, at least partial designs of the protective unit **40** by tarpaulins, films, textiles or the like are also conceivable. The netting receiving apparatus **12** has a fastening unit **50** for fastening to the construction machine **10**. The fastening unit **50** can also be seen in FIG. **6**, which shows a schematic perspective rear view of the netting receiving apparatus **12**.

[0059] FIG. **7** schematically shows the device **58** with a netting receiving apparatus **12**, the closure element **22** of which is equipped with an alternative inhibiting element **20'**. The alternative inhibiting element **20'** is partially designed as a roll **104**. In this case, the device **58** has a plurality of alternative inhibiting elements **20'** which are each designed as rolls **104**. The alternative inhibiting elements **20'** which are designed as rolls **104** are arranged in accordance with a half mesh extent **30** of the netting of the netting unit **14**.

[0060] FIG. **8** shows a schematic flow diagram of a method for providing nettings in mining and/or

tunnel construction by means of the system **56**. In at least one method step **106**, the device **58** is mounted on the boom **62** of the construction machine **10**. In at least one method step **108**, the closure elements **22**, **32** are moved into the maximum open position. In at least one method step **110**, the netting receiving apparatus **12** is moved analogously to an excavator blade in such a way that a netting unit **14** is received in the netting receiving space **34**. For this purpose, the netting insertion aid element **46** is moved under the netting unit **14** by a movement of the boom **62** and is subsequently tilted slightly. Alternatively, the netting unit **14** can also be manually inserted into the netting receiving space **34**. In at least one further method step **112**, the closure elements **22**, **32** are closed. In at least one further method step **114**, the netting is pulled out of the netting dispensing opening **18**. This can be done manually, for example. In this case, the inhibiting elements **20** are elastically bent, with the result that they exert a pressure on the netting. In at least one method step **60**, uncontrolled netting output is prevented by exerting the pressure on the netting. In at least one method step **116**, the netting is fastened to the rock wall **68** at a first point. This can be done, for example, with the aid of the further boom **64** of the construction machine **10**. In at least one further method step **118**, the device **58** is moved further along the rock wall **68** by the boom **62**. In this case, the netting, which is already fastened to the rock wall **68** at the first point, is pulled further out of the netting receiving apparatus **12**. In at least one further method step **120**, the further-pulled-out piece of the netting is fastened to a second point of the rock wall **68**, for example with the aid of the second boom **64**. The two last-mentioned method steps **118**, **120** are repeated multiple times until an entire netting web formed by the netting unit **14** is anchored to the rock wall **68**. In the favorable case, the netting web of the netting unit **14** extends over an entire length of a tunnel cross section/adit cross section.

Claims

1. A device for providing nettings by means of a construction machine, in particular in mining and/or tunnel construction, having at least one netting receiving apparatus which is configured to receive and at least partially enclose a netting unit and which has a netting receiving opening and a netting output opening, wherein at least the netting output opening is equipped with at least one inhibiting element which is configured to prevent uncontrolled netting output.
2. The device according to claim 1, the inhibiting element is of elastic design.
3. The device according to claim 1, wherein the inhibiting element is configured to apply a pressing to at least part of the netting.
4. The device according to claim 3, wherein the inhibiting element is configured to apply the pressing to at least that part of the netting which is currently emerging from the netting output opening.
5. The device according to claim 1, wherein the inhibiting element is arranged on the netting receiving apparatus in such a way that, at least in an unloaded state, the inhibiting element covers the netting output opening at least in sections at least for the most part.
6. The device according to claim 1, wherein the inhibiting element is replaceably attached to the netting receiving apparatus.
7. The device according to claim 1, wherein the netting receiving apparatus has at least one, in particular movably supported, closure element for at least partially closing the netting receiving opening.
8. The device according to claim 7, wherein the netting receiving apparatus has at least one guide rail which is configured to guide the closure element on a circumference of the netting receiving apparatus.
9. The device according to claim 7, wherein, in a closed state, the closure element delimits the netting output opening in sections.
10. The device at least according to claim 7, wherein the inhibiting element is configured to apply a

pressing to at least part of the netting, and wherein the closure element is configured to adjust the pressing applied by the inhibiting element.

11. The device according to claim 7, wherein the closure element is designed to be substantially narrower than the netting receiving opening in a main extension direction of the netting receiving opening.

12. The device at least according to claim 9, wherein the inhibiting element is arranged at a netting output opening-side end of the closure element.

13. The device according to claim 1, wherein the inhibiting element and/or the closure element has a width which corresponds to at least one mesh extent of the netting configured for provision.

14. The device according to claim 7, wherein the netting receiving apparatus has at least one further closure element which is in particular at least substantially identical to the closure element.

15. The device according to claim 1, wherein the netting receiving apparatus forms an at least substantially cylindrical netting receiving space which is configured to receive at least the netting unit.

16. The device according to claim 1, wherein the netting receiving apparatus has a netting receiving space which is overwidth, in particular in relation to web widths of netting units which can be provided by the netting receiving apparatus, and which is configured to provide a tolerance range for varying packing widths of, in particular otherwise identically designed, netting units.

17. The device according to claim 1, wherein at least one wall of the netting receiving apparatus has lightweight construction recesses.

18. The device according to claim 1, further comprising a protective unit which covers at least part of an outer side of the netting receiving apparatus in order to avoid injuries and/or soiling.

19. The device according to claim 1, wherein the netting receiving apparatus has a support element for positioning the netting receiving apparatus on a base.

20. The device according to claim 1, wherein at least one region of the netting receiving apparatus which delimits the netting receiving opening has a netting insertion aid element which is in particular configured to support an insertion of the netting unit into a netting receiving space of the netting receiving apparatus.

21. The device according to claim 1, wherein the netting receiving apparatus has a bearing unit for rotatably bearing netting roll central rods.

22. The device according to claim 1, wherein the netting receiving apparatus has a fastening unit for fastening to the construction machine.

23. The device according to claim 1, wherein the netting receiving apparatus has stiffening elements which extend parallel to a main extension direction of the netting receiving apparatus.

24. The device according to claim 1, wherein the inhibiting element is designed as a roll.

25. The device according to claim 24, further comprising a plurality of inhibiting elements which are designed as rolls and are arranged in accordance with a mesh extent or a half mesh extent of the netting unit configured for provision.

26. A system having the device according to claim 1 and having at least one netting unit, in particular a netting roll.

27. The system according to claim 26, having a construction machine configured for laying nettings, wherein the device is fixedly connected to the construction machine.

28. A method for providing nettings, in particular in mining and/or tunnel construction, by means of a device according to claim 1 wherein an uncontrolled netting output is prevented in at least one method step.
